

Applied Industrial Data Science

Week 2, Lecture 2: Problem Framing, Stakeholders, and Data Suitability

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WEEK 2 QUESTION

Before we train a model,
what decision must improve?

Good problem framing begins with a decision, not a dataset.

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Week 1 gave us the full decision system



Week 2 begins at the first link: define the problem before choosing data or a model.

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WEEK 2 FOCUS

Problem framing before
model selection

A clear problem statement guides every later choice.

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A useful request names a decision

✓ Vague request

Build an AI system to predict bread sales.

Missing: a user, an action, a deadline, and the cost of being wrong.

⊗ Decision-ready request

Help the FreshMart manager decide how many loaves to order for tomorrow.

*Added: the store manager, an order quantity, and "tomorrow".
Still missing: a success measure and limits.*

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Four questions create the first frame

👤 User

Who will use the output?

FreshMart: the store manager.

🎯 Decision

What choice may change?

FreshMart: how many loaves to order for tomorrow.

📖 Evidence

What information is needed?

FreshMart: sales, stockouts, promotions, calendar, weather.

⚠️ Consequence

What happens if we are wrong?

FreshMart: waste if too high; empty shelves if too low.

Your Week 1 problem statement asked exactly these four. This week we complete the frame around them.

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The Complete Problem Frame

Five parts should be clear before modelling begins.

👤 Who

Users, decision owners, affected people, and data owners.

🎯 What

The exact problem and decision to support.

🔗 How

The action that may follow from the output.

✓ Success

How useful improvement will be measured.

⚠️ Limits

Constraints, risks, and conditions for non-use.

If one part is unclear, the project is not ready for modelling.

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A stakeholder is not only the end user

👤 User

Sees or operates the output.

FreshMart: the manager reviewing tomorrow's order.

🎯 Decision owner

Authorises and owns the action.

FreshMart: the manager authorised to place or change the order.

📖 Affected people

Experience the benefit or harm.

FreshMart: customers, bakery staff, colleagues handling waste.

⚠️ Data owner

Controls access, quality, and use.

FreshMart: the retailer that holds the sales and stock records.

One person can hold two roles. That never removes the other two.

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Ask what each stakeholder needs from the system

Value

What outcome matters to them?

Manager: fewer empty shelves. Finance: less waste.

Evidence

What proof would build trust?

A comparison with last week's order — not a model score.

Risk

What failure worries them?

Being blamed for a skip of bread, or for a Saturday stockout.

Authority

Who may act, review, or stop use?

The manager can override; the retailer can withdraw the data.

Conflicting answers are not a problem to hide. They are the trade-off the project must make visible.

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DECISION STATEMENT

For [USER], use [EVIDENCE]
to support [DECISION].

Add the success measure, action window, and main constraint.

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INDUSTRIAL CASE

FreshMart problem frame before modelling

User, decision, and evidence

User: FreshMart store manager.

Decision: how many loaves to order for tomorrow.

Evidence: sales, stockouts, promotions, weekday, and weather.



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Success has more than one meaning

Technical

The forecast estimates tomorrow's demand well.

Operational

The result arrives before the decision.

Guardrail

Waste and stockouts do not increase.

Adoption

The manager understands and uses it.

Two more complete the set: can the recommendation actually be acted on, and does it beat the current rule?

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Set success criteria before modelling

Area	Example question
Technical	How close is the forecast to demand?
Operational	Is it ready before the order deadline?
Actionability	Can the suggested quantity be supplied and stored?
Guardrail	Do waste or stockouts increase?
Adoption	Does the manager review and use it?
Baseline	Is it better than last week's order or the current rule?

Write these as questions now. Targets come later — from evidence or stakeholders, never invented.

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Four constraints shape the feasible solution



1 Data access

Sales, stock, promotions, and history.



2 Action window

The order is due before the shop closes.



3 People and budget

Storage and supplier pack sizes limit choices.



4 Safety and governance

The manager reviews the recommendation.

Constraints belong in the problem definition.

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False positives and false negatives have different costs



False positive

The system warns of high demand, but demand is normal.
Cost: too much bread and more waste.



False negative

The system misses a high-demand day.
Cost: empty shelves, lost sales, and unhappy customers.

Define the event first. Here the event is a high-demand day; without it, "false positive" means nothing.

A confusion matrix counts these errors. It cannot tell you which one matters more.

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INDUSTRIAL CASE

Case: which error is more costly?

Errors and the alert threshold

False alarm: too much bread may be ordered.

Missed warning: shelves may become empty.

The alert threshold balances waste and stockouts.



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Responsible AI begins with the problem frame

The same question, widened: not only what an error costs, but who pays it.



Affected people

Who may be harmed or excluded?

FreshMart: customers who find empty shelves; staff who handle waste. FreshMart: the shop pays for waste; customers pay for stockouts.



Error cost

Who bears false alarms or misses?



Data rights

Is the data use lawful and needed?

FreshMart: sales and stock only — no customer identities needed.



Human oversight

Who can review, challenge, or stop it?

FreshMart: the manager, if the recommendation arrives in time with a reason.

This is the first entry in your ethical risk statement — the one a capstone cannot pass without.

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DATASET REALITY

A dataset provides evidence for a defined problem.

First define the decision. Then ask whether the data can support it.

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Check data before project approval

Check	Question
Source and licence	Do we know the source, and may we use it?
Data status	Is it public, simulated, synthetic, restricted — or unknown?
Sensitive data	Does it contain personal or confidential fields?
Question fit	Can its variables, unit of observation and time span answer the question?
Outcome availability	Is the outcome we want to predict actually recorded?
Scale	Is it manageable within one semester?
Permission	Is an approval, agreement or ethics review required?
Cross-border and confidentiality	International client, cross-border transfer, or a confidentiality duty?

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Cross-border work and confidentiality

Situation	What you must check
Pakistani law	PECA 2016, data-protection rules, and university ethics approval.
Personal data	Personal or human-subject fields need consent and clearance first.
International work	UK and EU GDPR, HIPAA for US health data, transfer limits.
Client and platform	Processing agreements, NDAs, platform terms, dataset licences.
Freelancing	Confidentiality outlives the contract. Coursework is no exemption.
If it is unclear	Ask before collecting. Unverified permission is a Pilot or a Stop.

The course gives no legal advice. What is assessed is recognising when permission is required before data is used.

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Data review leads to one of four decisions

Proceed

Suitable, permitted, and manageable.

FreshMart: sales, stock and stockouts are recorded and licensed for use.

Pilot

Continue carefully while uncertainty is checked.

FreshMart: proceed while confirming that stockouts are actually logged.

Change question

Use the data for a narrower decision.

FreshMart: sales without stock levels → forecast sales, not true demand.

Stop

Illegal, unsafe, unethical, or infeasible.

FreshMart: loyalty-card identities requested without consent.

Change and Stop are analytical conclusions, not failures. They earn full credit when evidenced.

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INDUSTRIAL CASE

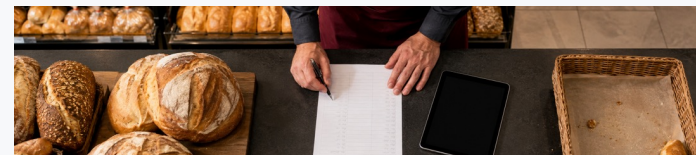
Final frame: tomorrow's FreshMart bread order

Decision, success, and limits

Decide tomorrow's bread order before the shop closes.

Reduce waste and stockouts compared with the current rule.

Verdict: Pilot. Manager review required while stockout logging is confirmed.



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This week's lab: same method, a different floor

Frame element	FreshMart (lecture)	Plant shift log (lab)
Request	"Predict bread sales."	"Use AI to predict machine problems."
User	Store manager	Maintenance supervisor
Decision	Tomorrow's bread order	Which line to inspect first
Action window	Before the shop closes	The next shift handover
Evidence	Sales, stock, promotions, calendar	Output, downtime, rejects, energy, motor temp (1 day)
Open question	Are stockouts recorded?	Is any failure or inspection outcome recorded?

Your four documents answer the right-hand column. Do not train a model this week.

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Four connected outputs complete the Week 2 practical

1 Problem brief

One page: user, decision, success, limits.

[docs/problem_brief.md](#)

2 Ethical risk note

Affected people, error costs, and oversight.

[docs/ethical_risk_note.md](#)

3 Dataset choice

Preliminary source, licence, and fit.

[docs/data_choice.md](#)

4 Suitability checklist

Feasibility, permission, scale, and risk.

[docs/data_suitability_checklist.md](#)

Version-track all four plus the executed notebook, then push. Week 1 commits stayed local; this week they reach the remote.

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EXIT REFLECTION

Can you explain your project without naming an algorithm?

Name the user, decision, success measure, main risk, and data needed.

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Q&A and Next Steps

Before the lab

Draft the one-page problem brief

Record assumptions and open questions

Commit and push the Week 2 outputs

Bring your Week 1 problem statement — you will upgrade it

Looking ahead

Week 3: Data Quality and Validation

Bring the dataset source and licence

Dataset approval checkpoint by end of Week 3

Provisional dataset choices get a first look in the lab

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